ALTERNATING CURRENT

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. AC BASICS: MEAN & RMS VALUES

$$i = I_0 \sin(\omega t + \phi), \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

$$I_{\text{rms}} = \sqrt{\langle i^2 \rangle} = \frac{I_0}{\sqrt{2}} \approx 0.707\, I_0$$

$$\bar{I}_{\text{half cycle}} = \frac{2I_0}{\pi} \approx 0.637\, I_0, \qquad \bar{I}_{\text{full cycle}} = 0$$

Form factor $= \dfrac{I_{\text{rms}}}{\bar{I}} = \dfrac{\pi}{2\sqrt{2}} \approx 1.11$

Indian mains: $V_{\text{rms}} = 220$ V $\Rightarrow V_0 \approx 311$ V, $f = 50$ Hz

Waveform	I_{rms}	Mean (full cycle)
Sinusoidal	$I_0/\sqrt{2}$	0
Half-wave rectified	$I_0/2$	I_0/π
Full-wave rectified	$I_0/\sqrt{2}$	$2I_0/\pi$
Square (symmetric)	I_0	0
Triangular / sawtooth	$I_0/\sqrt{3}$	0 (symm.)

Useful results:

- $i = a\sin\omega t + b\cos\omega t \;\Rightarrow\; I_0 = \sqrt{a^2+b^2}$
- $i = I_1 + I_0\sin\omega t \;\Rightarrow\; I_{\text{rms}} = \sqrt{I_1^2 + \frac{I_0^2}{2}}$

⋆ Common Pitfall

- AC ammeters/voltmeters read **rms** values; "220 V" is rms
- Mean over a full cycle is zero — always state the interval
- rms of a sum ≠ sum of rms values; for different frequencies add squares: $I^2_{\text{rms}} = I^2_{1,\text{rms}} + I^2_{2,\text{rms}}$

2. PURE R, L AND C

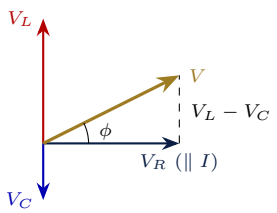
	R	L	C
Opposition	R	$X_L = \omega L$	$X_C = \dfrac{1}{\omega C}$
Phase	in phase	V leads I by $\frac{\pi}{2}$	I leads V by $\frac{\pi}{2}$
X vs f	const.	$\propto f$	$\propto 1/f$
DC ($f=0$)	R	short (0)	open (∞)
High f	R	open	short
P_{avg}	$I^2_{\text{rms}}R$	0	0

Memory aid: **ELI the ICE man** — in L, emf before current; in C, current before emf.

⋆ Common Pitfall

- X_L and X_C have unit ohm but dissipate no power
- "Leads by $\pi/2$" means the peak occurs $T/4$ earlier

3. SERIES LCR CIRCUIT



$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \qquad I_0 = \frac{V_0}{Z}$$

$$\tan\phi = \frac{X_L - X_C}{R}, \qquad \cos\phi = \frac{R}{Z}$$

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

- $X_L > X_C$: inductive (V leads I) $\quad X_C > X_L$: capacitive
- RL: $Z = \sqrt{R^2 + \omega^2 L^2}$ $\quad$ RC: $Z = \sqrt{R^2 + \dfrac{1}{\omega^2 C^2}}$

⋆ Common Pitfall

- Voltages add as **phasors**, not algebraically: $V \neq V_R + V_L + V_C$
- V_L or V_C alone can be **greater** than the source voltage
- Same current flows through all three elements in series

4. RESONANCE

$$\omega_0 = \frac{1}{\sqrt{LC}}, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

At resonance: $X_L = X_C$, $Z = R$ (min), $I_0 = \dfrac{V_0}{R}$ (max), $\phi = 0$, $\cos\phi = 1$

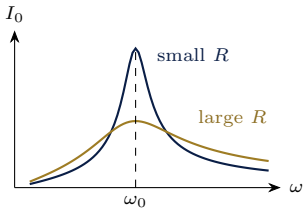
Quality factor:

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{\omega_0}{\Delta\omega}$$

Bandwidth: $\Delta\omega = \omega_2 - \omega_1 = \dfrac{R}{L}, \quad \Delta f = \dfrac{R}{2\pi L}$

Half-power points: $\omega_{1,2} \approx \omega_0 \mp \dfrac{R}{2L}$

At resonance: $V_L = V_C = Q\,V$



⋆ Common Pitfall

- $\omega < \omega_0$: circuit is capacitive; $\omega > \omega_0$: inductive
- ω_0 does not depend on R; sharpness does (smaller $R \Rightarrow$ sharper)
- Ideal parallel LC at resonance: $Z \to \infty$, line current minimum (rejector circuit)

5. POWER IN AC CIRCUITS

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi = I_{\text{rms}}^2 R$$

Power factor: $\cos\phi = \dfrac{R}{Z}$

Apparent power $S = V_{\text{rms}} I_{\text{rms}}$ (VA)

Reactive power $Q = V_{\text{rms}} I_{\text{rms}} \sin\phi$ (VAR), $S^2 = P^2 + Q^2$

Wattless current: $I_{\text{rms}} \sin\phi$ (component $\perp$ to V)
Active current: $I_{\text{rms}} \cos\phi$

⋆ Common Pitfall

- Power is dissipated **only in** R; pure L or C $\Rightarrow P = 0$
- Choke coil (large L, tiny R) limits current with almost no power loss — better than a resistor
- Instantaneous power $p = vi$ oscillates at 2ω

6. LC OSCILLATIONS

$$q = q_0 \cos\omega t, \quad i = -q_0\omega \sin\omega t, \quad \omega = \frac{1}{\sqrt{LC}}$$

$$I_0 = q_0\omega = \frac{q_0}{\sqrt{LC}}$$

$$\frac{q^2}{2C} + \frac{1}{2}Li^2 = \frac{q_0^2}{2C} = \frac{1}{2}LI_0^2$$

$U_E = U_B$ when $q = q_0/\sqrt{2}$, i.e. first at $t = T/8$

LC	q	i	L,	$1/C$
Spring–mass	x	v	m,	k

⋆ Common Pitfall

- Charge/current oscillate at f; energy (U_E, U_B) oscillates at $2f$
- With resistance, oscillations are damped (energy lost as heat)

7. AC GENERATOR & TRANSFORMER

Generator: $\varepsilon = NBA\omega \sin\omega t, \quad \varepsilon_0 = NBA\omega$

Ideal transformer:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = k, \qquad V_p I_p = V_s I_s$$

Efficiency: $\eta = \dfrac{P_{\text{out}}}{P_{\text{in}}} = \dfrac{V_s I_s}{V_p I_p}$

Energy loss	**Reduced by**
Copper (I^2R)	Thick copper wires
Eddy current	Laminated core
Hysteresis	Soft iron core (narrow loop)
Flux leakage	Winding coils one over other

Transmission: $P_{\text{loss}} = \left(\dfrac{P}{V}\right)^2 R_{\text{line}} \Rightarrow$ transmit at high voltage

⋆ Common Pitfall

- Step-up raises voltage but lowers current; power is NOT increased
- Frequency is unchanged by a transformer
- Transformer does not work on steady DC (no changing flux)

8. QUICK REVISION TABLE

Circuit	Z	ϕ (V **w.r.t.** I)	$\cos\phi$
R	R	0	1
L	ωL	$+\pi/2$	0
C	$1/\omega C$	$-\pi/2$	0
RL	$\sqrt{R^2+\omega^2L^2}$	$\tan^{-1}\frac{\omega L}{R}$	R/Z
RC	$\sqrt{R^2+\frac{1}{\omega^2C^2}}$	$-\tan^{-1}\frac{1}{\omega CR}$	R/Z
LCR	$\sqrt{R^2+(X_L-X_C)^2}$	$\tan^{-1}\frac{X_L-X_C}{R}$	R/Z

9. JEE SHORTCUTS & MASTER TIPS

1. $I_{\text{rms}} = I_0/\sqrt{2}$; half-cycle mean $= 2I_0/\pi$
2. Draw the phasor triangle: V_R along I, $(V_L - V_C)$ perpendicular
3. Resonance $\Rightarrow Z = R$, pf $= 1$, $V_L = V_C = QV$
4. $\Delta\omega = R/L$, $Q = \omega_0/\Delta\omega$
5. Two readings given (V_R, V_L, V_C) $\Rightarrow V = \sqrt{V_R^2 + (V_L - V_C)^2}$
6. Time from zero to peak $= T/4$; from zero to rms value $= T/8$

⋆ Master Confusion List

1. **rms vs peak:** meters show rms; insulation is designed for peak
2. **Phase sign:** $\phi > 0$ means V leads I (inductive)
3. X_L **vs** X_C **with** f**:** $X_L \uparrow$, $X_C \downarrow$ as $f \uparrow$
4. **Power:** $P = V_{\text{rms}} I_{\text{rms}} \cos\phi$, not $V_{\text{rms}} I_{\text{rms}}$
5. **Bandwidth:** $\Delta f = R/2\pi L$ (not $R/\pi L$)
6. **Wattless current** is $I_{\text{rms}} \sin\phi$, not $I_{\text{rms}} \cos\phi$

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